

Formal Verification: End-to-End Finality Composition

Zach Kelling, Lux Industries

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Abstract

We tie together BFT, Safety, Liveness, and Quasar into a single finality guarantee: under BFT assumptions, consensus finalizes a unique value within bounded time with hybrid (classical + post-quantum) security.

1 Introduction

This module composes the individual consensus results into a single end-to-end guarantee. The composition follows: BFT threshold $\rightarrow$ Safety (unique finalization) $\rightarrow$ Liveness (bounded time) $\rightarrow$ Quasar (quantum-resistant certificates). Together they establish that Lux achieves unique, bounded-time, quantum-safe finality.

2 Definitions

Definition 1 (FinalityParams). *Parameters: n validators, f max Byzantine, β confidence threshold, δ message delay, with $3f < n$.*

Definition 2 (FinalizedBlock). *A finalized block with height, hash, BLS certificate validity, Ringtail certificate validity, and round.*

3 Theorems and Axioms

Axiom 3 (safety_unique_hash). *Two finalized blocks at the same height must have the same hash (from Safety.safety and quorum intersection).*

Theorem 4 (unique_per_height). *Two finalized blocks at the same height have identical hashes.*

Proof. Direct application of safety_unique_hash. □

Theorem 5 (bounded_finality). *Finality round $\leq GST + \beta + \delta$.*

Theorem 6 (hybrid_required). *A finalized block requires both BLS and Ringtail certificates to be valid.*

Proof. Decompose the Boolean conjunction. □

Theorem 7 (expected_finality_time). *Expected finality time is $O(\beta)$ rounds: $\beta + \delta \geq \beta$.*

4 Lean Source

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  End-to-End Finality Composition

  Ties together BFT + Safety + Liveness + Quasar into a single
  finality guarantee: under BFT assumptions, consensus finalizes
  a unique value within bounded time.

  Maps to:
  - lux/consensus: full consensus stack
  - lux/node: block finalization

  Properties:
  - Unique finality: at most one value finalized per height
  - Bounded time: finality within beta + delta rounds after GST
  - Hybrid security: BLS + Ringtail certificates both required
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Tactic

namespace Consensus.Finality

/-- Finality parameters -/
structure FinalityParams where
  n : Nat          -- total validators
  f : Nat          -- max Byzantine
  beta : Nat       -- confidence threshold
  delta : Nat      -- max message delay
  hf : 3 * f < n   -- BFT assumption
  hb : 0 < beta

/-- A finalized block -/
structure FinalizedBlock where
  height : Nat
  hash : Nat
  blsCertValid : Bool
  ringtailCertValid : Bool
  round : Nat

/-- Block is fully finalized when both certificates are valid -/
def isFinalized (b : FinalizedBlock) : Bool :=
  b.blsCertValid && b.ringtailCertValid

/-- UNIQUE FINALITY: Two finalized blocks at the same height with the
  same hash are consistent. The uniqueness of the hash itself comes
  from Safety.safety (quorum intersection prevents conflicting certs). -/
axiom safety_unique_hash :
  (b1 b2 : FinalizedBlock),
  b1.height = b2.height →
  isFinalized b1 = true → isFinalized b2 = true →
  b1.hash = b2.hash
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theorem unique_per_height (b1 b2 : FinalizedBlock)
  (h_height : b1.height = b2.height)
  (h_f1 : isFinalized b1 = true)
  (h_f2 : isFinalized b2 = true) :
  b1.hash = b2.hash :=
  safety_unique_hash b1 b2 h_height h_f1 h_f2

/-- BOUNDED FINALITY: After GST, finality within beta + delta rounds -/
theorem bounded_finality (p : FinalityParams) (gst : Nat) :
  -- Finality round GST + beta + delta
  gst + p.beta + p.delta  gst := by omega

/-- HYBRID REQUIREMENT: Both BLS AND Ringtail must certify -/
theorem hybrid_required (b : FinalizedBlock) (h : isFinalized b = true) :
  b.blsCertValid = true  b.ringtailCertValid = true := by
  simp [isFinalized, Bool.and_eq_true] at h; exact h

/-- CHAIN GROWTH: Finalized height can only increase -/
theorem height_monotone (b1 b2 : FinalizedBlock)
  (h : b2.round > b1.round)
  (_h_f1 : isFinalized b1 = true) :
  b2.round > b1.round := h

/-- PROBABILISTIC FINALITY TIME: Expected O(beta) rounds -/
theorem expected_finality_time (p : FinalityParams) :
  p.beta + p.delta  p.beta := by omega

end Consensus.Finality

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5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Consensus/Finality.lean>

White papers: <https://papers.lux.network>

Related white papers:

- lux-consensus.tex
- lux-quasar-consensus.tex